BadgeOverlay

User Guide

v2026.05.13.001

Accessibility-focused badge & overlay tool for screen reader assistance

Windows 10/11 • .NET 10 • WPF/MVVM

1. Overview

BadgeOverlay is a lightweight accessibility tool that places visual badges over UI elements detected by Windows UI Automation (UIA), MSAA, or a custom rule engine. It is designed for users who rely on screen readers or need enhanced visual cues to navigate complex desktop UIs.

Key capabilities: real-time element detection, configurable info overlays, category-colour coding, Screen Reader Mode, 5 saveable presets, global hotkeys, and full WCAG 2.2 AAA compliance (7:1 contrast, 44 px touch targets, 2 px Gold/Cyan focus rings).

2. System Requirements

Component	Requirement
OS	Windows 10 v1903 or Windows 11
Runtime	.NET 10 Desktop Runtime (x64)
RAM	64 MB minimum
Display	1024×768 minimum, DPI-aware
Accessibility	UI Automation service must be enabled
Permissions	Standard user — no elevation required

3. Installation

3.1 First-Time Install

Run the BadgeOverlay installer (.exe or .msi). The installer places binaries in %LOCALAPPDATA%\BadgeOverlay and creates a Start Menu shortcut. No administrator rights required.

3.2 Portable Mode

Copy the publish folder to any location and run BadgeOverlay.exe directly. Settings are stored in %APPDATA%\BadgeOverlay\settings.json.

3.3 Updating

Close the application, run the new installer, and restart. Presets and settings are preserved across updates. Refer to the Update Manual for version-specific migration steps.

4. Interface Tour

The UI is divided into five functional zones:

Zone	Location	Purpose
1 Identity/Scale	Top bar	App name, version, DPI scale selector (100/125/150/200%)
2 NavRow	Row 2	Primary navigation: Detection | Overlays | Colours | Hotkeys | Presets | Settings

3 Toggle/Preview	Row 3	Master enable toggle + live preview of active badge styles
4 Main / Sliders	Centre	Active panel content — sliders, dropdowns, rule lists, colour pickers
5 Presets 1–5 / Save	Bottom bar	Quick-access preset slots + Save button; resize grip at bottom-right

■ All interactive targets are ≥44 px per WCAG 2.5.5. Focus rings are 2 px Gold (Ctrl+Shift+B) or Cyan (presets).

5. Quick Start

1. Launch BadgeOverlay from the Start Menu or taskbar shortcut.
2. Click the master toggle (Zone 3) or press Ctrl+Shift+B to enable overlay.
3. Mouse over any desktop element — a badge appears showing element type, name, and role.
4. Open Detection (NavRow) to choose detection engine priority: UIA → MSAA → Custom Rules.
5. Open Colours to assign category colours; open Overlays to configure info fields.
6. Press Ctrl+Shift+S to save the current configuration to the active preset slot.

6. Detection Engine

6.1 Engine Priority

BadgeOverlay queries element information in a cascading pipeline. Each stage falls through if the previous returns insufficient data:

Priority	Engine	Description
1	UI Automation (UIA)	Native Windows accessibility API. Highest fidelity — returns Name, ControlType, AutomationId, IsK
2	MSAA	Legacy Active Accessibility. Used when UIA returns empty or unsupported elements.
3	Custom Rule Engine	User-defined rules matching by window class, process name, or bounding rect. Highest flexibility fo

6.2 Custom Rules

Navigate to Detection → Custom Rules. Click Add Rule to open the rule editor. Rules match on one or more of: Window Class (exact or wildcard), Process Name, Control Type, Bounding Rect (relative percentages). Assign a display name, category, and colour to matching elements.

```
Example rule: ProcessName=game*.exe + ControlType=Custom → Category=HUD Element, Colour=#00d4ff
```

■ Rules are evaluated top-to-bottom. Drag to reorder. Rules earlier in the list take precedence.

7. Screen Reader Mode

Screen Reader Mode integrates with NVDA, JAWS, Narrator, and other AT tools. When enabled, BadgeOverlay feeds element metadata through the active screen reader's virtual buffer on hover, without requiring the element to receive keyboard focus.

Enable via: Overlays → Screen Reader Mode toggle, or press Ctrl+Shift+B to cycle modes.

Mode	Behaviour

Off	No screen reader announcements
Hover Announce	Reads element name + role on hover (500 ms dwell)
Focus Track	Announces on keyboard focus change only
Both	Hover + Focus Track combined

8. Info Overlay

The Info Overlay is a floating tooltip-style panel that displays rich element metadata. Configure which fields appear in Overlays → Info Fields:

Field	Source	Default
Element Name	UIA.Name / MSAA.accName	✓ Visible
Control Type	UIA.ControlType	✓ Visible
Automation ID	UIA.AutomationId	✓ Visible
Bounding Rect	UIA.BoundingRectangle	Hidden
Is Keyboard Focusable	UIA.IsKeyboardFocusable	Hidden
Process Name	Win32 PID lookup	Hidden
MSAA Role	MSAA.accRole	Hidden
Custom Rule Match	Rule engine	✓ Visible

9. Category Colours

Elements are assigned to categories. Each category has an independent border colour, fill opacity, and label colour. Open Colours in the NavRow to manage categories.

Default categories and colours:

Category	Default Colour	Notes
Button	#ffd700 Gold	Primary interactive elements
Input	#00d4ff Cyan	Text fields, combo boxes, sliders
Label	#a0a0b8 Dim	Static text, group labels
Container	#bd93f9 Purple	Panels, group boxes, tab pages
Image	#50fa7b Green	Image controls, icons
Custom Rule	User-defined	Assigned per rule

10. Global Hotkeys

Hotkey	Action
Ctrl+Shift+B	Toggle overlay on/off
Ctrl+Shift+F	Freeze current badge (hold overlay in place)
Ctrl+Shift+C	Copy element info to clipboard
Ctrl+Shift+S	Speak element info via screen reader
Ctrl+Shift+1–5	Load preset 1–5
Ctrl+Shift+P	Cycle to next preset

■ Hotkeys are registered globally and work even when BadgeOverlay is minimised. Customise in Settings → Hotkeys.

11. Presets

BadgeOverlay stores 5 named presets. Each preset captures: detection engine priority, active categories and colours, overlay fields, Screen Reader Mode, and all hotkey assignments.

Zone 5 (bottom bar) shows preset slots 1–5. Click a slot to load it. Click Save (or Ctrl+Shift+S) to write the current configuration into the active slot. Right-click a slot to rename it.

■ Presets are stored in %APPDATA%\BadgeOverlay\presets.json and can be copied between machines.

12. Settings Reference

Setting	Type	Default	Description
UiScale	Enum	1.0	1.0/1.25/1.5/2.0 — DPI multiplier for all UI elements
ThemeName	String	Dark	One of 20 built-in themes
ThemeAuto	Bool	false	Follow Windows light/dark mode
BadgeOpacity	Float	0.85	0.1–1.0 — overall badge transparency
OverlayDwell	Int	500	ms before Info Overlay appears
FreezeKey	String	Ctrl+Shift+F	Hotkey to freeze current badge
ScreenReaderMode	Enum	Off	Off/Hover/Focus/Both
ShowBoundingRect	Bool	false	Draw bounding rect outline on hover
LogLevel	Enum	Warning	Debug/Info/Warning/Error

13. Themes

BadgeOverlay includes 20 built-in themes selectable from the Identity bar or via Alt+T:

Theme	Style

Dark	Default — deep navy, Gold/Cyan accents
Light	Off-white, charcoal text
Nord	Arctic blues
Dracula	Purple/pink on dark
Catppuccin Mocha	Warm dark
Catppuccin Latte	Warm light
Ayu Dark	Orange accents
Ayu Light	Bright orange
Ayu Mirage	Muted dark
Cobalt2	Electric blue
GitHub Light	Developer grey
Gruvbox Dark	Earthy warm
Monokai	Classic neon
One Dark	Atom-inspired
Rose Pine	Muted rose
Solarized Dark	Ethan Schoonover dark
Solarized Light	Ethan Schoonover light
Tokyo Night	Purple night
Warm Light	Cream/amber
High Contrast	WCAG 7:1 guaranteed

14. Accessibility Compliance

Criterion	Level	Implementation
1.4.3 Contrast (min)	AA	All text ≥4.5:1
1.4.6 Contrast (enhanced)	AAA	All text ≥7:1
2.5.5 Target Size	AA	All targets ≥44×44 px
2.4.7 Focus Visible	AA	2 px Gold ring on keyboard focus
1.4.1 Use of Colour	A	No colour-only information; labels always present
1.3.1 Info & Relationships	A	GroupBox labels, radio button groups properly marked
2.1.1 Keyboard	A	All functions accessible via keyboard

1.2.3 Reduce Motion	AA	Animations disabled when Reduce Motion is set in Windows

15. Troubleshooting

Symptom	Likely Cause	Fix
No badges appear	Master toggle is off	Press Ctrl+Shift+B or click the toggle in Zone 3
Badges show 'Unknown'	UIA disabled or blocked app	Enable UIA in Windows accessibility settings; try MSAA fallback
High CPU usage	Custom rule polling too fast	Increase rule poll interval in Settings → Performance
Overlay flickers	GPU compositing conflict	Disable hardware acceleration in Settings → Display
Screen reader not reading	AT not detected	Launch AT before BadgeOverlay; check SR Mode setting
Hotkeys not working	Conflict with another app	Reassign hotkeys in Settings → Hotkeys

16. Log Files

Logs are written to: %APPDATA%\BadgeOverlay\logs\badgeoverlay_YYYY-MM-DD.log

Log level is configurable (Settings → Log Level). Default: Warning. For support, set to Debug, reproduce the issue, then attach the log file.

```
Log path: C:\Users\\AppData\Roaming\BadgeOverlay\logs\
```

17. Build Notes

Version v2026.05.13.001 was built with .NET 10 SDK, C# 14 language features (field keyword), and WPF. JSON serialisation uses System.Text.Json source generation for AOT compatibility. Versioning follows the yyyy.MM.dd.NNN scheme; the build number auto-increments and is synchronised across the .csproj, build pipeline, and NSI installer script.

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